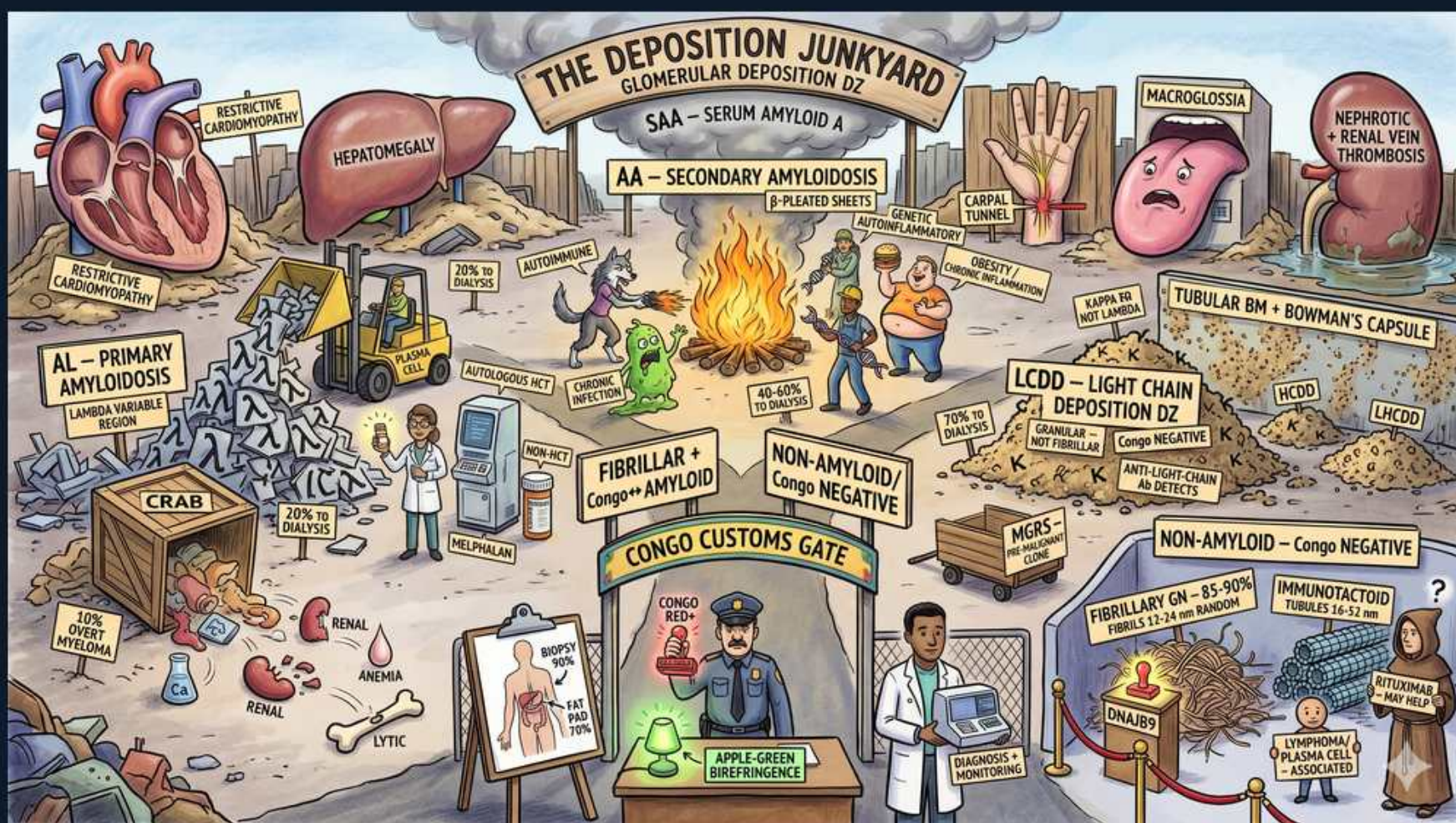


Glomerular Deposition Diseases — Amyloid, MIDD & Fibrillary



1. AL — primary amyloidosis

WHAT YOU SEE

AL primary amyloidosis: lambda light-chain (clonal plasma cells), melphalan

WHAT IT ENCODES

AL amyloidosis is deposition of monoclonal immunoglobulin LIGHT CHAINS (lambda > kappa) from a plasma-cell dyscrasia. Renal involvement causes nephrotic-range proteinuria. Workup: serum/urine free light chains and SPEP/UPEP; treat the clone (e.g., bortezomib/daratumumab-based, autologous transplant, historically melphalan).

BOARD TRAP

AL = light-chain (lambda) amyloid from plasma cells; AA = serum amyloid A from chronic inflammation. Both are Congo-red positive — distinguish by typing (mass spec/IHC).

2. AA — secondary amyloidosis

WHAT YOU SEE

AA secondary amyloidosis: serum amyloid A, beta-pleated sheets, chronic inflammation

WHAT IT ENCODES

AA (secondary) amyloidosis deposits serum amyloid A protein, an acute-phase reactant from chronic inflammation (RA, IBD, chronic infection, familial Mediterranean fever). Like all amyloid it forms beta-pleated sheets. Treat the underlying inflammatory driver.

BOARD TRAP

AA is from chronic inflammation/FMF, not a plasma-cell clone — light chains are normal; don't chase a myeloma workup as the cause.

3. Congo red — apple-green birefringence

WHAT YOU SEE

Congo customs gate: apple-green birefringence under polarized light

WHAT IT ENCODES

All amyloid stains with Congo red and shows pathognomonic APPLE-GREEN birefringence under polarized light. EM shows randomly arranged 8–12 nm non-branching fibrils. This is the defining test that separates amyloid (Congo+) from non-amyloid deposition diseases.

BOARD TRAP

Apple-green birefringence under POLARIZED light is the key — Congo red alone (red staining) without polarization is not diagnostic.

4. Fibrillar + Congo+ = amyloid

WHAT YOU SEE

Sign: fibrillar + Congo+ → amyloid pathway

WHAT IT ENCODES

The deposition-disease algorithm splits on Congo red. Fibrillar deposits that are Congo-red POSITIVE = amyloid (AL/AA). Fibrillar deposits that are Congo-red NEGATIVE = fibrillary GN. Granular non-fibrillar deposits = MIDD (light/heavy chain deposition disease).

BOARD TRAP

Fibrils on EM do NOT automatically mean amyloid — fibrillary GN is also fibrillar but Congo-red NEGATIVE.

5. LCDD — light chain deposition

WHAT YOU SEE

LCDD light chain deposition dz: granular, kappa, anti-light-chain on IF, non-amyloid Congo negative

WHAT IT ENCODES

Monoclonal immunoglobulin deposition disease (MIDD) — most often LIGHT-chain deposition disease (LCDD), usually KAPPA — deposits GRANULAR (non-fibrillar) material along basement membranes. Congo red is NEGATIVE; IF shows monotypic light chain. Causes nodular glomerulosclerosis resembling Kimmelstiel–Wilson.

BOARD TRAP

LCDD is KAPPA-predominant and Congo-NEGATIVE; AL amyloid is LAMBDA-predominant and Congo-POSITIVE — opposite light chain, opposite Congo.

6. Nodular GS mimics KW

WHAT YOU SEE

Tubular BM + Bowman's capsule deposits, nodular glomerulosclerosis

WHAT IT ENCODES

LCDD produces nodular mesangial sclerosis that mimics diabetic Kimmelstiel–Wilson nodules on light microscopy and also deposits along tubular basement membranes and Bowman's capsule. The distinction is made by Congo-red negativity and monotypic light-chain IF.

BOARD TRAP

Nodular glomerulosclerosis without diabetes/retinopathy — think LCDD, not diabetic nephropathy; check light chains.

7. Fibrillary GN — 10–30 nm

WHAT YOU SEE

Fibrillary GN: fibrils 10-30nm, Congo negative, IgG/DNAJB9

WHAT IT ENCODES

Fibrillary GN deposits randomly arranged fibrils larger than amyloid (~10–30 nm) that are Congo-red NEGATIVE, with polyclonal IgG and a 'smudgy' IF. DNAJB9 is a sensitive/specific marker. Presents with proteinuria, hematuria, and progressive renal failure.

BOARD TRAP

Fibrillary GN fibrils are THICKER (10–30 nm) and Congo NEGATIVE; amyloid fibrils are THINNER (8–12 nm) and Congo POSITIVE.

8. Immunotactoid GN — microtubules

WHAT YOU SEE

Immunotactoid: microtubules >30nm, CLL/lymphoma associated

WHAT IT ENCODES

Immunotactoid glomerulopathy deposits larger, organized MICROTUBULAR structures (>30 nm) often in parallel arrays, typically monoclonal, and is associated with CLL/lymphoproliferative disorders. Screen for an underlying hematologic malignancy.

BOARD TRAP

Immunotactoid = thick microtubules >30 nm + lymphoproliferative association; fibrillary GN = thinner randomly arranged fibrils, usually no malignancy.

9. MGRS

WHAT YOU SEE

MGRS — monoclonal gammopathy of renal significance, treat the clone

WHAT IT ENCODES

When a small B-cell/plasma-cell clone that does NOT meet myeloma criteria still causes renal injury (LCDD, AL amyloid, etc.), it is termed monoclonal gammopathy of renal significance (MGRS). Despite 'benign' hematologic burden, clone-directed therapy is indicated to save the kidney.

BOARD TRAP

MGRS is treated like a malignancy (clone-directed therapy) even though it does NOT meet myeloma diagnostic criteria — 'MGUS-level' clone, but renally significant.

10. CRAB — myeloma features

WHAT YOU SEE

CRAB: hyperCalcemia, Renal, Anemia, Bone lytic lesions

WHAT IT ENCODES

Multiple myeloma end-organ damage = CRAB: hyperCalcemia, Renal failure, Anemia, Bone (lytic) lesions. Myeloma kidney is most often cast nephropathy from filtered light chains, distinct from glomerular deposition diseases.

BOARD TRAP

Cast nephropathy (myeloma kidney) is TUBULAR obstruction by light-chain casts — different from glomerular AL amyloid/LCDD even though all stem from a plasma-cell clone.

11. Cardiac restrictive (amyloid)

WHAT YOU SEE

Restrictive cardiomyopathy + hepatomegaly + macroglossia (AL amyloid systemic)

WHAT IT ENCODES

Systemic AL amyloid deposits in the heart (restrictive cardiomyopathy, low-voltage ECG with thick walls), liver (hepatomegaly), tongue (macroglossia), and nerves (carpal tunnel). These extrarenal clues point to amyloid over MIDD.

BOARD TRAP

Macroglossia and bilateral carpal tunnel are classic for AL amyloid systemic deposition — not features of diabetic nodular sclerosis or LCDD.

12. Nephrotic + renal vein thrombosis

WHAT YOU SEE

Nephrotic syndrome + renal vein thrombosis risk

WHAT IT ENCODES

Heavy proteinuria from glomerular deposition disease produces nephrotic syndrome with urinary loss of antithrombin III and a hypercoagulable state, predisposing to renal vein thrombosis (also classically seen in membranous nephropathy).

BOARD TRAP

Renal vein thrombosis with nephrotic syndrome is most classic for membranous nephropathy, but any heavy nephrotic state (incl. amyloid) raises the risk.
